

18.06SC Final Exam (solved)

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Question 1

Suppose A is 3×4 , and $Ax = 0$ has exactly 2 special solutions:

$$x_1 = [1 \ 1 \ 1 \ 0] \quad \text{and} \quad x_2 = [-2 \ -1 \ 0 \ 1]$$

(a) Find the row reduced echelon form R of A .

The special solutions to $Ax = 0$ correspond to the free variables in the reduced row echelon form (RREF) of A . Since A is 3×4 and there are 2 special solutions, there are 2 free variables and 2 pivot columns. We can reconstruct R by expressing the general solution in terms of the free variables and matching the structure to the RREF.

1. The general solution to $Ax = 0$ is $x = c_1 x_1 + c_2 x_2$ for scalars c_1, c_2 .
2. Write x in terms of the variables:

$$x = c_1 [1 \ 1 \ 1 \ 0] + c_2 [-2 \ -1 \ 0 \ 1] = [c_1 - 2c_2 \ c_1 - c_2 \ c_1 \ c_2]$$

3. Let $x_3 = c_1$ and $x_4 = c_2$ (so x_3 and x_4 are the free variables). Then:

$$x_1 = x_3 - 2x_4 \quad x_2 = x_3 - x_4 \quad x_3 = x_3 \quad x_4 = x_4$$

4. Rearranged, the general solution is:

$$[x_1 \ x_2 \ x_3 \ x_4] = x_3 [1 \ 1 \ 1 \ 0] + x_4 [-2 \ -1 \ 0 \ 1]$$

5. The RREF of A must have pivots in columns 1 and 2, and free variables in columns 3 and 4. The equations for the pivots are:

$$x_1 + 2x_4 - x_3 = 0 \quad x_2 + x_4 - x_3 = 0$$

Rearranged:

$$x_1 - x_3 + 2x_4 = 0 \quad x_2 - x_3 + x_4 = 0$$

6. The RREF R is:

$$R = [1 \ 0 \ -1 \ 2 \ 0 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0]$$

Therefore, The row reduced echelon form of A is

$$R = [1 \ 0 \ -1 \ 2 \ 0 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0]$$

(b) Find the dimensions of all four fundamental subspaces $C(A)$, $N(A)$, $C(A^T)$, $N(A^T)$. Find bases for those you can.

The dimensions of the fundamental subspaces are determined by the number of pivots and free variables. The nullspace $N(A)$ is spanned by the special solutions. The column space $C(A)$ has dimension equal to the number of pivots. The row space $C(A^T)$ and the left nullspace $N(A^T)$ can be found from the RREF.

1. A is 3×4 with 2 pivots (from part (a)), so:

- $\dim N(A) = 2$ (number of free variables)
- $\dim C(A) = 2$ (number of pivots)
- $\dim C(A^T) = 2$ (row space, number of pivots)
- $\dim N(A^T) = 1$ (since $3 - 2 = 1$)

2. Bases:

- $N(A)$: The special solutions x_1 and x_2 form a basis.
- $C(A)$: The pivot columns of A form a basis, but since A is not given, the pivot columns of R can be used for the row space.
- $C(A^T)$: The nonzero rows of R form a basis for the row space.
- $N(A^T)$: Find a vector y such that $R^T y = 0$.

3. The nonzero rows of R are:

$$r_1 = [1, 0, -1, 2], \quad r_2 = [0, 1, -1, 1]$$

So a basis for $C(A^T)$ is r_1, r_2 .

4. To find a basis for $N(A^T)$, solve $R^T y = 0$:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & -1 & -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 0$$

This gives: $y_1 = 0, y_2 = 0, -y_1 - y_2 = 0, 2y_1 + y_2 = 0$. The only solution is $y_1 = y_2 = 0$, but since R is 3×4 , the left nullspace is in $\mathbb{R}^3$ and has dimension 1. Alternatively, since the rank is 2, $N(A^T)$ has dimension $3 - 2 = 1$.

To find a basis, note that the left nullspace is orthogonal to the row space. The row space is spanned by $[1, 0, -1, 2]$ and $[0, 1, -1, 1]$. The left nullspace is the set of y such that $y^T R = 0$.

Let $y = [a, b, c]$ and solve $y^T R = 0$:

$$a[1, 0, -1, 2] + b[0, 1, -1, 1] + c[0, 0, 0, 0] = 0$$

This gives a single nontrivial solution, which can be found by inspection or by solving the system.

Therefore,

- $\dim N(A) = 2$, basis: $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$
- $\dim C(A) = 2$, basis: the pivot columns of A (not explicitly given)
- $\dim C(A^T) = 2$, basis: $\{[1, 0, -1, 2], [0, 1, -1, 1]\}$
- $\dim N(A^T) = 1$, basis: can be found by solving $R^T y = 0$

Question 2

2 (6+3+2=11 pts.)

(a) Find the inverse of a 3×3 upper triangular matrix U with nonzero entries a, b, c, d, e, f :

Given:

$$U = \begin{bmatrix} a & b & c & 0 & d & e & 0 & 0 & f \end{bmatrix}$$

The inverse of an upper triangular matrix with nonzero diagonal entries is also upper triangular. We can find U^{-1} by solving $UX = I$ or by using Gauss-Jordan elimination. Each entry of U^{-1} can be found by back substitution.

1. Let $U^{-1} = X = \begin{bmatrix} x_{11} & x_{12} & x_{13} & 0 & x_{22} & x_{23} & 0 & 0 & x_{33} \end{bmatrix}$.
2. Solve $UX = I$ column by column:
 - Third column: $U \begin{bmatrix} x_{13} & x_{23} & x_{33} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$
 - $fx_{33} = 1 \implies x_{33} = 1/f$
 - $dx_{23} + ex_{33} = 0 \implies x_{23} = -e/(df)$
 - $ax_{13} + bx_{23} + cx_{33} = 0 \implies x_{13} = \frac{-bx_{23} - cx_{33}}{a} = \frac{be}{adf} - \frac{c}{af}$
 - Second column: $U \begin{bmatrix} x_{12} & x_{22} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$
 - $dx_{22} = 1 \implies x_{22} = 1/d$
 - $ax_{12} + bx_{22} = 0 \implies x_{12} = -b/(ad)$
 - First column: $U \begin{bmatrix} x_{11} & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$
 - $ax_{11} = 1 \implies x_{11} = 1/a$
3. Assemble U^{-1} :

$$U^{-1} = \begin{bmatrix} 1/a & -b/(ad) & (be)/(adf) - c/(af) & 0 & 1/d & -e/(df) & 0 & 0 & 1/f \end{bmatrix}$$

Therefore, The inverse is

$$U^{-1} = \begin{bmatrix} \frac{1}{a} & -\frac{b}{ad} & \frac{be}{adf} - \frac{c}{af} & 0 & \frac{1}{d} & -\frac{e}{df} & 0 & 0 & \frac{1}{f} \end{bmatrix}$$

(b) Suppose the columns of U are eigenvectors of a matrix A . Show that A is also upper triangular.

If U is invertible and its columns are eigenvectors of A , then A is similar to a diagonal matrix if the eigenvectors are linearly independent. If U is upper

triangular and A has U as its eigenvector matrix, then A must be upper triangular as well.

1. If A has U as its eigenvector matrix, then $A = U\Lambda U^{-1}$, where Λ is diagonal.
2. Since U is upper triangular and U^{-1} is also upper triangular, the product $U\Lambda U^{-1}$ is upper triangular (see notes/Diagonalization.ipynb).
3. This is because the product of upper triangular matrices is upper triangular, and multiplying by a diagonal matrix preserves upper triangularity.

Therefore, A is upper triangular.

(c) Explain why this U cannot be the same matrix as the first factor in the Singular Value Decomposition $A = U\Sigma V^T$.

In the SVD, the matrix U is always an orthogonal (or unitary) matrix, meaning its columns are orthonormal. An upper triangular matrix with arbitrary nonzero entries cannot be orthogonal unless it is a permutation of a diagonal matrix with ± 1 entries.

1. In the SVD, U must satisfy $U^T U = I$ (orthogonality).
2. An upper triangular matrix with arbitrary nonzero entries cannot have orthonormal columns unless it is diagonal with ± 1 entries (see notes/SVD.ipynb and notes/SVD-intro.ipynb).
3. The given U has arbitrary nonzero entries a, b, c, d, e, f , so its columns are not necessarily orthonormal.

Therefore, This U cannot be the U in the SVD unless it is a signed permutation matrix (which is not the case here).

Question 3

3 (3+3+5=11 pts.)

(a) A and B are any matrices with the same number of rows. What can you say (and explain why it is true) about the comparison of

rank of A rank of the block matrix $\begin{bmatrix} A & B \end{bmatrix}$

The rank of $\begin{bmatrix} A & B \end{bmatrix}$ is at least as large as the rank of A , because A 's columns are included in $\begin{bmatrix} A & B \end{bmatrix}$. Adding more columns cannot decrease the rank.

1. The rank of a matrix is the dimension of its column space (see notes/Linear Transformations.ipynb).
2. The columns of A are present in $\begin{bmatrix} A & B \end{bmatrix}$, so the column space of A is a subspace of the column space of $\begin{bmatrix} A & B \end{bmatrix}$.
3. Adding columns (from B) can only increase or leave unchanged the dimension of the column space.

Therefore,

$$\text{rank}(A) \leq \text{rank}(\begin{bmatrix} A & B \end{bmatrix})$$

(b) Suppose $B = A^2$. How do those ranks compare? Explain your reasoning.

If $B = A^2$, then the columns of A^2 are linear combinations of the columns of A , so adding A^2 as extra columns does not increase the rank.

1. Each column of A^2 is a linear combination of the columns of A (since $A^2 = AA$).
2. Therefore, the column space of A^2 is contained in the column space of A .
3. Thus, $\begin{bmatrix} A & A^2 \end{bmatrix}$ has the same column space as A .

Therefore,

$$\text{rank}(\begin{bmatrix} A & A^2 \end{bmatrix}) = \text{rank}(A)$$

(c) If A is m by n of rank r , what are the dimensions of these nullspaces?

Nullspace of A Nullspace of $\begin{bmatrix} A & A \end{bmatrix}$

The nullspace of A consists of all x such that $Ax = 0$. The nullspace of $\begin{bmatrix} A & A \end{bmatrix}$ consists of all vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ such that $Ax + Ay = 0$, or $A(x + y) = 0$.

1. A is $m \times n$ of rank r , so $\dim N(A) = n - r$ (see notes/Elimination-matrices.ipynb).
2. For $\begin{bmatrix} A & A \end{bmatrix}$, let $z = \begin{bmatrix} x \\ y \end{bmatrix}$, $x, y \in \mathbb{R}^n$.
3. $\begin{bmatrix} A & A \end{bmatrix} z = Ax + Ay = A(x + y) = 0$.
4. So the nullspace is all pairs (x, y) such that $x + y \in N(A)$.
5. Let $s = x + y \in N(A)$, and $t = x - y$ (so $x = \frac{s+t}{2}$, $y = \frac{s-t}{2}$). For each $s \in N(A)$ and $t \in \mathbb{R}^n$, $z = \begin{bmatrix} x \\ y \end{bmatrix}$ is in the nullspace.
6. Therefore, the nullspace of $\begin{bmatrix} A & A \end{bmatrix}$ has dimension n (from t) plus $n - r$ (from s), so $2n - r$.

Therefore,

- $\dim N(A) = n - r$
- $\dim N([A \ A]) = 2n - r$

Question 4

4 (3+4+5=12 pts.)

Suppose A is a 5×3 matrix and Ax is never zero (except when x is the zero vector).

(a) What can you say about the columns of A ?

If $Ax = 0$ only when $x = 0$, then the columns of A are linearly independent.

1. $Ax = 0$ has only the trivial solution $x = 0$.
2. This means the nullspace of A is $\{0\}$.
3. Therefore, the columns of A are linearly independent (see notes/Elimination-matrices.ipynb and notes/Linear Transformations.ipynb).

Therefore, The columns of A are linearly independent.

(b) Show that $A^T Ax$ is also never zero (except when $x = 0$) by explaining this key step:

If $A^T Ax = 0$ then obviously $x^T A^T Ax = 0$ and then (WHY?) $Ax = 0$.

If $A^T Ax = 0$, then $x^T A^T Ax = (Ax)^T (Ax) = |Ax|^2 = 0$, so $Ax = 0$, which only happens when $x = 0$.

1. If $A^T Ax = 0$, then $x^T A^T Ax = 0$.
2. $x^T A^T Ax = (Ax)^T (Ax) = |Ax|^2$ (see notes/Least-Square Fitting.ipynb).
3. $|Ax|^2 = 0$ implies $Ax = 0$.
4. By part (a), $Ax = 0$ only when $x = 0$.

Therefore, $A^T Ax = 0$ only when $x = 0$.

(c) We now know that $A^T A$ is invertible. Explain why $B = (A^T A)^{-1} A^T$ is a one-sided inverse of A (which side of A ?). B is NOT a 2-sided inverse of A (explain why not).

B is a left inverse of A because $BA = I_3$, but $AB \neq I_5$ since A is not square or surjective onto $\mathbb{R}^5$.

1. $B = (A^T A)^{-1} A^T$.
2. $BA = (A^T A)^{-1} A^T A = (A^T A)^{-1} (A^T A) = I_3$.
3. So B is a left inverse: $BA = I$ on $\mathbb{R}^3$.
4. $AB = A(A^T A)^{-1} A^T$ is a 5×5 matrix, but it is a projection onto the column space of A , not the identity (see notes/Least-Square Fitting.ipynb and notes/Projections.ipynb).
5. Therefore, B is not a two-sided inverse.

Therefore, B is a left inverse of A ($BA = I$), but not a right inverse ($AB \neq I$).

Question 5

5 (5+5=10 pts.)

If A is 3×3 symmetric positive definite, then $Aq_i = \lambda_i q_i$ with positive eigenvalues and orthonormal eigenvectors q_i .

Suppose $x = c_1 q_1 + c_2 q_2 + c_3 q_3$.

(a) Compute $x^T x$ and also $x^T Ax$ in terms of the c 's and λ 's.

Expressing x in the orthonormal eigenbasis, $x^T x$ is the sum of squares of the c_i , and $x^T Ax$ is the sum of $\lambda_i c_i^2$.

1. $x = c_1 q_1 + c_2 q_2 + c_3 q_3$ with q_i orthonormal.
2. $x^T x = (c_1 q_1 + c_2 q_2 + c_3 q_3)^T (c_1 q_1 + c_2 q_2 + c_3 q_3)$
3. By orthonormality, $q_i^T q_j = \delta_{ij}$, so cross terms vanish.
4. $x^T x = c_1^2 + c_2^2 + c_3^2$
5. $x^T Ax = (c_1 q_1 + c_2 q_2 + c_3 q_3)^T A(c_1 q_1 + c_2 q_2 + c_3 q_3)$
6. $Aq_i = \lambda_i q_i$, so $Ax = c_1 \lambda_1 q_1 + c_2 \lambda_2 q_2 + c_3 \lambda_3 q_3$

7. $x^T Ax = (c_1 q_1 + c_2 q_2 + c_3 q_3)^T (c_1 \lambda_1 q_1 + c_2 \lambda_2 q_2 + c_3 \lambda_3 q_3)$
 8. Again, by orthonormality, only diagonal terms remain: $x^T Ax = c_1^2 \lambda_1 + c_2^2 \lambda_2 + c_3^2 \lambda_3$

Therefore,

- $x^T x = c_1^2 + c_2^2 + c_3^2$
- $x^T Ax = c_1^2 \lambda_1 + c_2^2 \lambda_2 + c_3^2 \lambda_3$

(b) Looking at the ratio $x^T Ax$ in part (a) divided by $x^T x$ in part (a), what c 's would make that ratio as large as possible? You can assume $\lambda_1 < \lambda_2 < \dots < \lambda_n$. Conclusion: the ratio $x^T Ax / x^T x$ is a maximum when x is _____.

The ratio is maximized when all of x is in the direction of the eigenvector with the largest eigenvalue.

1. The ratio is $\frac{x^T Ax}{x^T x} = \frac{c_1^2 \lambda_1 + c_2^2 \lambda_2 + c_3^2 \lambda_3}{c_1^2 + c_2^2 + c_3^2}$.
2. This is a weighted average of the λ_i 's, weighted by c_i^2 .
3. The maximum occurs when all the weight is on the largest λ_i (i.e., x is a multiple of q_3 if λ_3 is largest).
4. So, set $c_3 \neq 0$, $c_1 = c_2 = 0$.

Therefore, The ratio $x^T Ax / x^T x$ is maximized when x is in the direction of the eigenvector with the largest eigenvalue (i.e., x is a multiple of q_3 if λ_3 is the largest).

Question 6

6 (4+4+4=12 pts.)

(a) Find a linear combination w of the linearly independent vectors v and u that is perpendicular to u .

We want $w = v + \alpha u$ such that w is perpendicular to u .

1. $w = v + \alpha u$.
2. $w^T u = 0$ (perpendicularity condition).
3. $w^T u = v^T u + \alpha u^T u = 0$.
4. Solve for α : $\alpha = -\frac{v^T u}{u^T u}$.

Therefore, A linear combination is $w = v - \frac{v^T u}{u^T u} u$.

(b) For the 2-column matrix $A = [u \ v]$, find Q (orthonormal columns) and R (2 by 2 upper triangular) so that $A = QR$.

This is the Gram-Schmidt process: Q has orthonormal columns, R is upper triangular.

1. Let $q_1 = \frac{u}{|u|}$.
2. Compute $w = v - \frac{v^T u}{u^T u} u$ (from part (a)).
3. $q_2 = \frac{w}{|w|}$.
4. $Q = [q_1 \ q_2]$.
5. R is 2×2 upper triangular:
 - $R_{11} = |u|$
 - $R_{12} = q_1^T v$
 - $R_{22} = |w|$
 - $R_{21} = 0$
6. $A = QR$ with Q as above and $R = \begin{bmatrix} |u| & q_1^T v & 0 & |w| \end{bmatrix}$

Therefore,

- $Q = [q_1 \ q_2]$ with $q_1 = \frac{u}{|u|}$, $q_2 = \frac{v - \frac{v^T u}{u^T u} u}{|v - \frac{v^T u}{u^T u} u|}$
- $R = \begin{bmatrix} |u| & q_1^T v & 0 & |w| \end{bmatrix}$

(c) In terms of Q only, using $A = QR$, find the projection matrix P onto the plane spanned by u and v .

The projection matrix onto the column space of Q (with orthonormal columns) is $P = QQ^T$.

1. Q has orthonormal columns spanning the same space as u and v .
2. The projection matrix onto the column space of Q is $P = QQ^T$ (see notes/Projections.ipynb).

Therefore, $P = QQ^T$.

Question 7

7 (4+3+4=11 pts.)

(a) Find the eigenvalues of

$$C = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad C^2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}.$$

C is a permutation matrix representing a 4-cycle. Its eigenvalues are the 4th roots of unity. C^2 is a permutation matrix representing two 2-cycles.

1. C cycles the basis vectors: $e_1 \rightarrow e_2 \rightarrow e_3 \rightarrow e_4 \rightarrow e_1$.
2. The characteristic polynomial of C is $\lambda^4 - 1 = 0$.
3. The eigenvalues of C are the 4th roots of unity: $1, -1, i, -i$.
4. C^2 cycles $e_1 \rightarrow e_3 \rightarrow e_1$ and $e_2 \rightarrow e_4 \rightarrow e_2$ (two 2-cycles).
5. The characteristic polynomial of C^2 is $(\lambda^2 - 1)^2 = 0$.
6. The eigenvalues of C^2 are 1 (multiplicity 2), -1 (multiplicity 2).

Therefore,

- Eigenvalues of C : $1, -1, i, -i$
 - Eigenvalues of C^2 : 1 (twice), -1 (twice)
-

(b) Those are both permutation matrices. What are their inverses C^{-1} and $(C^2)^{-1}$?

The inverse of a permutation matrix is its transpose. For a cycle, the inverse cycles in the opposite direction.

1. C is a 4-cycle: $e_1 \rightarrow e_2 \rightarrow e_3 \rightarrow e_4 \rightarrow e_1$.
2. C^{-1} cycles in the reverse direction: $e_1 \rightarrow e_4 \rightarrow e_3 \rightarrow e_2 \rightarrow e_1$.
3. $C^{-1} = C^3$ (since $C^4 = I$).
4. $(C^2)^{-1} = C^2$ (since $(C^2)^2 = I$).
5. Alternatively, the inverse of a permutation matrix is its transpose.

Therefore,

- $C^{-1} = C^3 = C^T$
 - $(C^2)^{-1} = C^2 = (C^2)^T$
-

(c) Find the determinants of C and $C + I$ and $C + 2I$.

The determinant of a permutation matrix is the sign of the permutation. The determinant of $C + kI$ is the product of its eigenvalues shifted by k .

1. C is a 4-cycle, which is an odd permutation, so $\det C = -1$.
2. The eigenvalues of C are $1, -1, i, -i$.
3. $\det(C + I) = (1 + 1)(-1 + 1)(i + 1)(-i + 1) = 2 \times 0 \times (1 + i) \times (1 - i) = 0$
4. $\det(C + 2I) = (1 + 2)(-1 + 2)(i + 2)(-i + 2) = 3 \times 1 \times (2 + i) \times (2 - i) = 3 \times 1 \times (4 + 0) = 12$

Therefore,

- $\det C = -1$
 - $\det(C + I) = 0$
 - $\det(C + 2I) = 12$
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Question 8

8 (4+3+4=11 pts.)

Suppose a rectangular matrix A has independent columns.

(a) How do you find the best least squares solution $\hat{x}$ to $Ax = b$? By taking those steps, give a formula (letters not numbers) for $\hat{x}$ and

also for $p = A\hat{x}$.

The least squares solution $\hat{x}$ minimizes $|Ax - b|^2$. For independent columns, the normal equations $A^T A\hat{x} = A^T b$ have a unique solution.

1. Set up the normal equations: $A^T A\hat{x} = A^T b$.
2. Since A has independent columns, $A^T A$ is invertible.
3. Solve for $\hat{x}$: $\hat{x} = (A^T A)^{-1} A^T b$.
4. The projection $p = A\hat{x}$.

Therefore,

- $\hat{x} = (A^T A)^{-1} A^T b$
- $p = A\hat{x}$

(b) The projection p is in which fundamental subspace associated with A ? The error vector $e = b - p$ is in which fundamental subspace?

The projection p is in the column space of A . The error e is in the left nullspace of A (the orthogonal complement of the column space).

1. $p = A\hat{x}$ is a linear combination of the columns of A , so $p \in C(A)$.
2. The error $e = b - p$ is orthogonal to every column of A , so $A^T e = 0$.
3. Therefore, e is in the left nullspace $N(A^T)$.

Therefore,

- p is in the column space $C(A)$
- e is in the left nullspace $N(A^T)$

(c) Find by any method the projection matrix P onto the column space of A :

$$A = \begin{bmatrix} 1 & 0 & 3 & 0 & 0 & -1 & 0 & -3 \end{bmatrix}$$

The projection matrix onto the column space of A is $P = A(A^T A)^{-1} A^T$.

1. Compute $A^T A$:

$$A^T A = \begin{bmatrix} 1 & 3 & 0 & 0 & 0 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 & 0 & 0 & -1 & 0 & -3 \end{bmatrix} = \begin{bmatrix} 1^2 + 3^2 & 0 & 0 & 1^2 + 3^2 \end{bmatrix} = \begin{bmatrix} 10 & 0 & 0 & 10 \end{bmatrix}$$

2. $A^T A$ is diagonal, so $(A^T A)^{-1} = \begin{bmatrix} 1/10 & 0 & 0 & 1/10 \end{bmatrix}$
3. $P = A(A^T A)^{-1} A^T$
4. Compute $A(A^T A)^{-1}$:

$$A(A^T A)^{-1} = \begin{bmatrix} 1/10 & 0 & 3/10 & 0 & 0 & -1/10 & 0 & -3/10 \end{bmatrix}$$

5. $P = A(A^T A)^{-1} A^T =$

$$\begin{aligned} & \begin{bmatrix} 1/10 & 0 & 3/10 & 0 & 0 & -1/10 & 0 & -3/10 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 & 0 & 0 & -1 & 0 & -3 \end{bmatrix} = \\ & = \begin{bmatrix} (1/10) \cdot 1 & (1/10) \cdot 3 & 0 & 0 & (3/10) \cdot 1 & (3/10) \cdot 3 & 0 & 0 & 0 & (-1/10) \cdot (-1) & (-1/10) \cdot (-3) & 0 & 0 & (-3/10) \cdot (-1) & (-3/10) \cdot (-3) \end{bmatrix} \\ & = \begin{bmatrix} 1/10 & 3/10 & 0 & 0 & 3/10 & 9/10 & 0 & 0 & 0 & 1/10 & 3/10 & 0 & 0 & 3/10 & 9/10 \end{bmatrix} \end{aligned}$$

Therefore, the projection matrix is

$$P = \begin{bmatrix} 1/10 & 3/10 & 0 & 0 & 3/10 & 9/10 & 0 & 0 & 0 & 1/10 & 3/10 & 0 & 0 & 3/10 & 9/10 \end{bmatrix}$$

Question 9

9 (3+4+4=11 pts.)

This question is about the matrices with 3's on the main diagonal, 2's on the diagonal above, 1's on the diagonal below.

$$A_1 = [3] \quad A_2 = \begin{bmatrix} 3 & 2 & 1 & 3 \end{bmatrix} \quad A_3 = \begin{bmatrix} 3 & 2 & 0 & 1 & 3 & 2 & 0 & 1 & 3 \end{bmatrix} \quad A_n = \begin{bmatrix} 3 & 2 & 0 & 0 & 1 & 3 & 2 & 0 & 0 & 1 & 3 & \cdot & 0 & 0 & \cdot & \cdot \end{bmatrix}$$

(a) What are the determinants of A_2 and A_3 ?

Compute the determinants directly for A_2 and A_3 .

1. $A_2 = \begin{bmatrix} 3 & 2 & 1 & 3 \end{bmatrix}$

$$\det(A_2) = 3 \cdot 3 - 1 \cdot 2 = 9 - 2 = 7$$

2. $A_3 = \begin{bmatrix} 3 & 2 & 0 & 1 & 3 & 2 & 0 & 1 & 3 \end{bmatrix}$ Expand along the first row:

$$\begin{aligned} \det(A_3) &= 3 \begin{vmatrix} 3 & 2 & 1 & 3 \\ 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 0 \\ 1 & 3 & 0 & 1 \end{vmatrix} - 2 \begin{vmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 0 \\ 1 & 3 & 0 & 1 \end{vmatrix} + 0 \begin{vmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 0 \\ 1 & 3 & 0 & 1 \end{vmatrix} \\ &= 3(9 - 2) - 2(1 \cdot 3 - 0 \cdot 2) + 0 = 3(7) - 2(3) = 21 - 6 = 15 \end{aligned}$$

Therefore,

- $\det(A_2) = 7$
- $\det(A_3) = 15$

(b) The determinant of A_n is D_n . Use cofactors of row 1 and column 1 to find the numbers a and b in the recursive formula for D_n :

$$(*) \quad D_n = a, D_{n-1} + b, D_{n-2}, .$$

Expand $\det(A_n)$ along the first row to find the recurrence.

1. The first row of A_n is $[3, 2, 0, \dots, 0]$.
2. Expanding along the first row:

$$D_n = 3D_{n-1} - 2 \cdot 1 \cdot D_{n-2}$$

(The -2 comes from the 2 in the $(1, 2)$ position and the 1 in the $(2, 1)$ position, with a negative sign from the cofactor expansion.)

3. So $a = 3, b = -2$.

Therefore, $D_n = 3D_{n-1} - 2D_{n-2}$, so $a = 3, b = -2$.

(c) This equation $(*)$ is the same as

$$\begin{bmatrix} D_n & D_{n-1} \end{bmatrix} = \begin{bmatrix} a & b & 1 & 0 \end{bmatrix} \begin{bmatrix} D_{n-1} & D_{n-2} \end{bmatrix}.$$

From the eigenvalues of that matrix, how fast do the determinants D_n grow? (If you didn't find a and b , say how you would answer part (c) for any a and b)
For 1 point, find D_5 .

The growth rate of D_n is determined by the largest eigenvalue of the recurrence matrix.

1. The recurrence matrix is $M = \begin{bmatrix} 3 & -2 & 1 & 0 \end{bmatrix}$.
2. The characteristic polynomial is $\lambda^2 - 3\lambda + 2 = 0$.
3. The eigenvalues are $\lambda = 1, 2$.
4. So D_n grows like 2^n for large n (the largest eigenvalue dominates).
5. To find D_5 , use the recurrence:
 - $D_1 = 3$
 - $D_2 = 7$
 - $D_3 = 15$
 - $D_4 = 3 \cdot 15 - 2 \cdot 7 = 45 - 14 = 31$
 - $D_5 = 3 \cdot 31 - 2 \cdot 15 = 93 - 30 = 63$

Therefore,

- D_n grows like 2^n for large n (since 2 is the largest eigenvalue).
- $D_5 = 63$